

Geography Ch-1 (Advanced) — India's Location & Extent

Advanced / India-applied | GS-I | Source: D.R. Khullar (India: A Comprehensive Geography)

Complements Must-Do Ch-1 (Earth & Universe) with the India-specific dimension

Note: world-physical basics are in the Must-Do folder

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

ADVANCED — India's Location, Extent & Time (D.R. Khullar)

■ NEWS TRIGGER

The recurring 'single time zone vs North-East time zone' debate applies the longitude-time concept to India — the sun rises ~2 hours earlier in Arunachal than in Saurashtra, yet all India uses one IST.

■ INTRO & ORIGIN

India lies wholly in the **Northern & Eastern hemispheres**, between about **8°4'N–37°6'N** and **68°7'E–97°25'E**. The mainland stretches ~3,214 km N–S and ~2,933 km E–W.

*The **Tropic of Cancer (23.5°N)** passes through the middle of India (8 states), dividing it into a tropical south and a sub-tropical north — a key control on India's climate.*

■ KEY DATA

Parameter	Value	Significance
Latitudinal extent	8°4'N – 37°6'N	Tropic of Cancer bisects India; climate contrast
Longitudinal extent	68°7'E – 97°25'E	~29° → ~2 hr sun-time gap E–W
IST meridian	82°30'E (via Mirzapur, UP)	One standard time for the whole country
Land / coastline	~3.28 M sq km; 7,516 km coast	7th-largest country

■ STATIC THEORY

- India's **latitudinal extent** gives it both tropical and sub-tropical characteristics; the Tropic of Cancer passes through 8 states (Gujarat to Mizoram).
- The ~29° **longitudinal spread** means a ~2-hour difference in local sun-time between the eastern and western extremes — resolved by adopting a single IST at 82°30'E.
- Despite the E–W time gap, India uses **one Indian Standard Time (UTC+5:30)** for administrative uniformity.

■ MUST-KNOW FACTS

- India: 8°4'N–37°6'N and 68°7'E–97°25'E.
- Tropic of Cancer (23.5°N) passes through 8 Indian states.
- IST meridian = 82°30'E (near Mirzapur); UTC+5:30.
- ~2-hour sun-time gap between Arunachal and Gujarat.

■ UPSC TRAPS

- ✗ The Equator passes through India.
- ✓ The TROPIC OF CANCER (23.5°N) passes through India, not the Equator.
- ✗ India has multiple official time zones.
- ✓ India uses a SINGLE IST (82°30'E, UTC+5:30) despite a ~2-hour E–W sun-time gap.

■ MAINS ANGLE

Use India's longitudinal extent to argue for/against a separate North-East time zone — energy savings and work-hours vs administrative and railway-safety complexity.

■ STUDY LINK

Geography → India → Location, Extent & Standard Time (Khullar Ch-1)

End of Register Notes — UPSC Agent / Copilot CLI